WORKSHEET

Binomial probability experiments

1 A binomial experiment has a probability of failure equal to 0.3. If there are five trials, then the probability of exactly one success is equal to:

A 0.0284

B 0.3602

C 0.0057

D 0.0085

E 0.0122

2 A binomial experiment has a random variable, X, equal to the number of successes after $n = 12$ trials. If each trial is independent of all others, and the probability of success is fixed at $p = 0.65$, then $P(X = 7)$ is given by:

A $\binom{7}{12} \times 0.65^7 \times 0.35^5$

B $\binom{12}{7} \times 0.65^7 \times 0.35^5$

C $\binom{12}{7} \times 0.35^7 \times 0.65^5$

D $\binom{12}{7} \times 0.65^7 \times 0.45^5$

E $\binom{12}{5} \times 0.65^5 \times 0.35^7$

3 Determine if the following are examples of binomial experiments. Circle the correct answer.

a Counting the number of coin flips required before a head appears. Yes/No

b Counting the number of coin flips required to get two successive tails. Yes/No

c Rolling a standard die 10 times and noting the number of times a score of five occurs. Yes/No

d Counting the number of randomly drawn cards from a standard deck until an ace is drawn. Yes/No

e Flipping a coin five times and counting the number of tails that occur. Yes/No

f Rolling two dice 20 times and recording the number of times a double score is observed. Yes/No

© Cengage Learning Australia Pty Ltd 2014

g Randomly drawing a card 10 times from a standard deck without replacement, and recording the number of hearts that appear. Yes/No

h Randomly drawing two cards from a standard deck 15 times, with replacement, and recording the number of times a pair occurs. Yes/No

4 A coin is flipped 8 times and random variable, X, is the number of heads observed.

a What is the value of n?

b What is the value of p?

c What is the domain of X?

d Does X have a binomial distribution?

5 A card is randomly drawn from a 52 card deck and noted before being replaced. The process is repeated 20 times. What is the probability that a queen appeared:

a zero times?

b on one occasion?

c on four occasions?

d on nine occasions?

6 A binomial variable, X, has a probability function given by

$$P(X = x) = \binom{6}{x} \times 0.72^{6-x} \times 0.28^{x} \text{ for } 0 \leq x \leq 6.$$

Find:

a the number of trials

© Cengage Learning Australia Pty Ltd 2014

b the domain of X

c the probability of success in any trial

d the probability distribution as a table (with probabilities correct to 4 d.p.).

7 A binomial variable, Y, has a probability function given by

$P(Y = y) = \binom{5}{y} \times 0.5^{y} \times 0.5^{5-y}$.

Draw a graph of the probability distribution of Y.

Answers

1 A

2 B

3 **a** No

b No

c Yes

d No

e Yes

f Yes

g No

h Yes

4 **a** 8

b 0.5

c $X = \{0,1,2,3,4,5,6,7,8\}$

d Yes

5 **a** ≈ 0.2017

b ≈ 0.3362

c ≈ 0.0471

d $\approx 6.566 \times 10^{-6}$

6 **a** 6

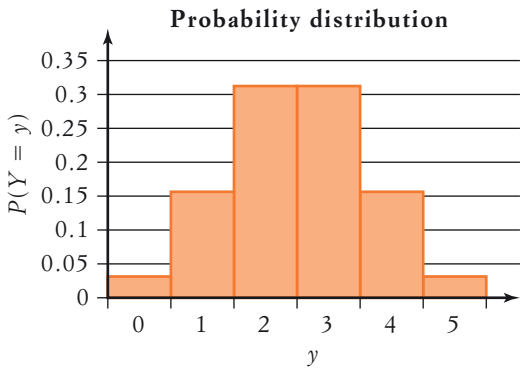
b $X = \{0,1,2,3,4,5,6\}$

c 0.28

d

x	0	1	2	3	4	5	6
$p(x)$	0.1393	0.3251	0.3160	0.1639	0.0478	0.0074	0.0005

7



© Cengage Learning Australia Pty Ltd 2014